SOFTWARE REQUIREMENTS SPECIFICATION · VERSION 3.0

Pothgula

A free and open-source web system for lending and membership in a small library operating across several branches.

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DATE
22 September 2026

STATUS
Draft for client review

This document replaces version 2.0 of 8 September 2026. Section 2 lists every change, with the number of the item in your list of sixteen beside each one. Requirement identifiers are kept from the earlier versions, so `FR-CIR-04` means the same thing in all three documents; withdrawn requirements keep their number and are shown struck through. Markers now show what changed since *version 2.0*; anything marked new or changed in v2.0 and untouched since is shown plain.

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CONTENTS

01 Purpose and scope

02 What changed since v2.0

03 Who uses the system

04 Accounts, registration and approval

05 Branches and categories

06 The book catalogue

07 Issuing, taking back and extending

08 Books from other branches

09 Lost and damaged books

10 Membership and annual fee

11 Opening hours

12 Notices and events

13 Alerts and reminders

14 The activity log

15 Reports and exports

16 Quality requirements

17 Information we store

18 What we recommend changing

19 Limitations and risks

20 Delivery in three phases

21 Decisions we need

22 Acceptance

01

Purpose and scope

Pothgula replaces the paper register. It keeps one shared record of every book, every member and every loan across all of your branches, tells your staff what needs attention today, and keeps a permanent record of who did what.

What the system does

- Holds the **catalogue** of every book copy the library owns, each with a number that says which branch accessioned it and which category it belongs to.
- Records **who borrowed what and when it is due back**, lets a member ask for **two more weeks**, and shows staff what is due soon and what is overdue.
- Lets a member **request a book held at another branch** and tracks that book on its way there and back.
- Settles **lost books** — the member either replaces the exact book or pays an amount staff decide, within 100 days.
- Takes **membership applications** from the public, holds them until a librarian approves, and tracks each membership year and its fee.
- Publishes each branch's **opening hours, notices and events** to members.
- Shows every warning and reminder **on the dashboard** of the person who needs to see it.
- Keeps an **activity log** of every action by every person, and exports it and every report as text and PDF.
- Produces **summaries** — most and least borrowed books, stock, fees collected, missing books.

What it does not do

This release is a lending and membership system, not a full library automation package. It does not do acquisitions and purchase orders, loans to or from other libraries, digital or e-book lending, online fee payment, or a public catalogue open to non-members. It also sends nothing outside itself — no SMS, no WhatsApp message, no email, no push notification. Section 19 explains the consequences of each of these; section 20 shows where they could be added later.

A CONSTRAINT THAT SHAPES EVERYTHING BELOW

Pothgula is a **free and open-source project**. Nothing in it may depend on a paid third-party service, or on a charge per message, per request or per user. That decision removed SMS reminders in version 2.0; in this version it decides how notice images and branch logos are stored (section 12) and rules out a WhatsApp gateway even though the WhatsApp number is now the sign-in name. It is written down as requirement NFR-11 so that it cannot be quietly reversed later.

HOW TO READ THIS DOCUMENT

Every requirement has an identifier such as `FR-CIR-04` and a priority. `MUST` means the system is not accepted without it. `SHOULD` means it is planned and will be dropped only if we agree to drop it. `COULD` means it is desirable and will be built if time allows. Requirements that are new or altered since version 2.0 are marked `NEW` or `CHANGED`; those removed are marked `WITHDRAWN` and greyed out. Please mark any requirement you disagree with before signing section 22.

02

What changed since version 2.0

Sixteen items in your list, which become twelve changes here because several belong together. The "your item" column gives the number from your list so you can check that nothing was missed. Each change is written out in full in the section named beside it.

YOUR ITEM	CHANGE	WHAT IT MEANS	WHERE
7, 8, 9	**Members are approved before they can borrow**	Registering from the public form now creates an *application*. The applicant can sign in and browse straight away but cannot borrow until an admin or super admin approves them. Staff see a list of waiting applications and can export any one person's details as a text file or a PDF. The registration form gains a WhatsApp number (required, and now the sign-in name), a second phone number, an organisation and the name of the person who introduced the member.	§4, §10
12, 13, 14	**A longer book number**	The accession number becomes `KS-A1-01234`: two letters for the branch, a letter and a digit for the category, five digits counting up. This lifts the ceiling from 26 branches to 676 and from 26 categories to 260, and answers question Q3 of v2.0.	§5, §6
15	**Branch name and logo**	A super admin can change a branch's name and the library's name at any time, and upload a logo for each branch that members see on their dashboard and that appears on printed exports. The two-letter code never changes, so renaming a branch does not touch a single book number.	§5

YOUR ITEM	CHANGE	WHAT IT MEANS	WHERE
5	**A loan can be extended by two weeks**	A member can ask for one extension of two weeks, counted from the original due date. An admin or super admin approves or declines it. This was a "could" in v2.0 (FR-CIR-10) and is now a "must" with a request-and-approve flow.	§7
3, 4	**Staff are warned five days ahead; late returns are called out**	Every loan due within five days appears on the staff dashboard with the member's details, the book and the branch. When a book comes back late, the counter screen says so plainly, the member's record keeps a count of late returns, and the member sees it on their own dashboard.	§7, §13
11	**Requesting a book from another branch**	A member borrows only at their home branch. For a book held elsewhere they submit a request; staff at both branches see the member's full details, the book travels to the home branch, is issued there, and travels back after return. This answers question Q1 of v2.0 and adds a small tracking flow that did not exist before.	§8
6, 16	**Lost books: staff set the amount, and the member has 100 days**	The book price is no longer required. When a book is declared lost, the admin or super admin decides the amount owed (the system suggests twice the price where one is recorded). The member then has 100 days to pay or bring the same book, and during that time may borrow one book instead of two. What happens on day 101 is question Q1 in section 21.	§6, §9
1, 2	**Notices and events**	An admin or super admin can publish a notice or an event — a title, an image and a paragraph, with a date and place for events — that members see on their dashboard. Images are stored inside the system, not on a free image website.	§12
10	**Logs and reports export as text and PDF**	Every log view and every report can be saved as a plain text file and as a PDF, alongside the spreadsheet export already in v2.0. All three are produced by the system itself.	§14, §15
—	**The annual fee is set by a super admin**	From your notes: the fee amount per branch is a setting that only a super admin can change. Written into FR-MEM-06.	§10
—	**Already covered**	Your note that admins and super admins can change a member's password is FR-AUT-11 and FR-AUT-12 of v2.0, unchanged. Barcode and QR labels and looking up book details by ISBN, which your notes mark for research, are FR-CAT-10, FR-CAT-16 and FR-CAT-17 in section 6 and are placed in phase 3.	§4, §6

Questions version 2.0 asked that are now answered

ASKED IN V2.0	ANSWER TAKEN INTO V3.0
Q1	A member borrows **only at their home branch** and **requests** a book held elsewhere; the book travels. Section 8.
Q2	The public form stays and produces an **application** that staff approve. Section 4.
Q3	**Two letters** for the branch, a letter and a digit for the category. Section 5.
Q5	**100 days** to replace or pay for a lost book. Section 9.
Q11	The lost-book charge is **decided by staff** when the book is declared lost; the recorded price is a suggestion only. Section 9.

Questions Q4, Q6 to Q10 and Q12 to Q15 from version 2.0 are still open. They are carried into section 21 together with the new ones raised by these changes.

03

Who uses the system

Three kinds of people use Pothgula, plus one temporary state: an applicant who has registered but not yet been approved. Everyone signs in with their WhatsApp number and a password.

MEMBER

The person who borrows books

Applies once, is approved by a librarian, pays an annual fee, and belongs to one home branch.

- Browses the catalogue of every branch and sees what is on the shelf
- Borrows at their home branch; requests books from other branches
- Asks for a two-week extension on a book they hold
- Sees due dates, what they owe, opening hours, notices and events in one panel
- Changes their own password and their own contact details

ADMIN · LIBRARIAN

Runs the counter at one branch

Day-to-day staff. Everything they do is limited to their own branch, and they cannot delete.

- Approves membership applications; issues books and takes them back
- Decides extension requests and handles other-branch requests
- Adds and edits books, registers members, publishes notices and events
- Records annual fees and lost-book settlements, and sets the lost-book amount
- Sets opening hours; reads and exports the branch's log

SUPER ADMIN

Runs the library organisation

Everything an admin can do, across every branch, plus the things that must not be undone by mistake.

- Creates branches with their two-letter codes, sets branch names and logos, defines categories
- Creates staff accounts; sets the annual fee for each branch
- Deletes records, waives lost-book charges, moves books between branches permanently
- Reads the activity log and the summaries for the whole library

Who can do what

CAPABILITY	APPLICANT	MEMBER	ADMIN	SUPER ADMIN
Browse the catalogue of any branch, see book details	**Yes**	**Yes**	**Yes**	**Yes**
See opening hours, notices and events	**Yes**	**Yes**	**Yes**	**Yes**
NEW Borrow a book	—	**Home branch**	—	—
NEW Request a book held at another branch	—	**Yes**	—	—
NEW Request a two-week extension	—	**Yes**	—	—
NEW Approve or reject a membership application	—	—	**Own branch**	**Any branch**
NEW Export one member's details (text, PDF)	—	—	**Own branch**	**Any branch**
Issue a book to a member	—	—	**Own branch**	**Any branch**
Take a book back	—	—	**Own branch**	**Any branch**
NEW Approve or decline an extension	—	—	**Own branch**	**Any branch**
NEW Accept, dispatch and receive other-branch requests	—	—	**Own branch**	**Any branch**
Correct a loan record	—	—	**Own branch**	**Any branch**

CAPABILITY	APPLICANT	MEMBER	ADMIN	SUPER ADMIN
Add or edit a book	—	—	**Own branch**	**Any branch**
Register a member at the counter (active at once)	—	—	**Own branch**	**Any branch**
Record an annual fee payment	—	—	**Own branch**	**Any branch**
CHANGED Declare a book lost and set the amount owed	—	—	**Own branch**	**Any branch**
Waive or reduce a lost-book charge	—	—	—	**Yes**
NEW Publish, edit and withdraw notices and events	—	—	**Own branch**	**Any branch or all**
Set opening days and times	—	—	**Own branch**	**Any branch**
Change own password	**Yes**	**Yes**	**Yes**	**Yes**
Reset another person's password	—	—	**Own branch**	**Anyone**
CHANGED Read and export the activity log (text, spreadsheet, PDF)	—	—	**Own branch**	**All branches**
Run summaries and reports	—	—	**Own branch**	**All branches**
NEW Change the annual fee	—	—	—	**Yes**
NEW Change a branch's name or logo, or the library name	—	—	—	**Yes**
NEW Create categories and their codes	—	—	—	**Yes**
Delete a book, loan, member, payment or any other record	—	—	—	**Yes**
Move a book to another branch permanently	—	—	—	**Yes**
Create a staff account	—	—	—	**Yes**
Create a branch and give it its two-letter code	—	—	—	**Yes**

THE APPLICANT COLUMN

An applicant is not a fourth role. It is a member whose application has not yet been approved. They have the same account, the same sign-in and the same dashboard; the only difference is that everything to do with borrowing is closed to them and a banner tells them why. The moment a librarian approves them, the banner goes and the borrowing features open.

Accounts, registration and approval

ID	REQUIREMENT	PRIORITY
FR-AUT-01	The landing page lets a visitor choose Member, Admin or Super admin, then choose to sign in or register.	MUST
FR-AUT-02	CHANGED Sign-in uses the person's **WhatsApp number** and password. Passwords are stored one-way hashed and can never be read back by anyone, including us.	MUST
FR-AUT-03	CHANGED A person applies for membership with: full name; address; WhatsApp number (required); another phone number (optional); date of birth (required); organisation, such as a school or workplace (optional); the name of the person who introduced them (required); and their home branch. Submitting the form creates an account in the **pending** state — not a membership.	MUST
FR-AUT-04	Registering as a super admin requires the pre-shared super admin code. At most five super admin accounts may exist per branch.	MUST
FR-AUT-05	CHANGED A super admin can create an account of any type at any branch. An admin can create a member account at their own branch. A member created by staff at the counter is **active immediately**: the librarian who created it is the approval.	MUST
FR-AUT-06	An account is locked for 15 minutes after five failed sign-in attempts. The same limit applies to attempts at the super admin code.	SHOULD
FR-AUT-07	A staff session ends after 30 minutes of inactivity, so a counter machine left unattended does not stay signed in.	SHOULD
~~FR-AUT-08~~	WITHDRAWN A member can reset a forgotten password using a code sent to their registered phone number. *Removed in v2.0. Still removed: a WhatsApp gateway is also a paid service.*	REMOVED
FR-AUT-09	A super admin can deactivate a staff or member account. Accounts are never deleted, so their loan history stays intact.	SHOULD
FR-AUT-10	Any signed-in person — applicant, member, admin or super admin — can change their own password by giving the current one and the new one twice. Changing a password ends every other session for that account.	MUST
FR-AUT-11	An admin can reset the password of any member or admin of their own branch. A super admin can reset the password of any account at any branch, including another super admin.	MUST
FR-AUT-12	A reset produces a temporary password shown once on screen for the staff member to hand over. The person must change it the first time they sign in and cannot use the system until they do. The old password is never shown and never recoverable.	MUST
FR-AUT-13	Every password change and every password reset is written to the activity log with who did it, to whom and when. The temporary password itself never appears in the log.	MUST
FR-AUT-14	At least two super admin accounts must exist at all times. The system refuses to deactivate the last two.	SHOULD
FR-AUT-15	A password must be at least eight characters, and the system refuses the most commonly used ones.	SHOULD
FR-AUT-16	NEW A **pending** applicant can sign in, browse the catalogue of every branch, open any book's details, and see opening hours, notices and events. They cannot be issued a book, request a book from another branch, or request an extension, and staff attempting to issue to them are told why. A banner on their dashboard says the application is waiting for approval at their home branch.	MUST

ID	REQUIREMENT	PRIORITY
FR-AUT-17	NEW An admin can approve or reject applications for their own branch; a super admin for any branch. Approval makes the account an active member and starts the membership year from that day (FR-MEM-01). Rejection requires a written reason; a rejected person can no longer sign in and is told at the sign-in screen to contact the branch. Both are logged.	MUST
FR-AUT-18	NEW The staff dashboard lists applications waiting for approval, oldest first, showing name, age (worked out from the date of birth), area (from the address), organisation, the person who introduced them, and the date applied. Opening one shows every detail given on the form. The number waiting appears in the notification panel.	MUST
FR-AUT-19	NEW From the pending list, and from any member's page, staff can export that **one person's** full details as a plain text file and as a PDF, each produced by the system itself with the branch name and logo on the PDF. Every export is logged with who exported whose details.	MUST
FR-AUT-20	NEW An application still pending after 30 days is flagged on the staff dashboard. Rejected applications, and pending ones the applicant never followed up, are removed after 90 days, because the library has no reason to keep the personal details of people who never became members. Both periods are settings.	SHOULD
FR-AUT-21	NEW An applicant can correct their own application while it is pending. A member can change their own address, other phone number and organisation at any time; changing the WhatsApp number, because it is the sign-in name, is done by staff at the counter.	SHOULD

WHAT THIS SETTLES, AND WHAT IT LEAVES

Approval closes the largest risk in version 2.0 — an unknown person on the internet creating a live member and walking out with books (risk D2). The public form stays, so nobody has to type at the counter what the applicant can type at home; the librarian's job becomes checking and pressing Approve. What is not yet decided is whether the **first year's fee** is taken at the moment of approval (question Q6 in section 21) and who exactly the **"person who introduced them"** is — a guarantor, a referee, or simply a note (Q5).

STILL OPEN FROM VERSION 1.0

FR-AUT-04 means the super admin code is the single thing standing between the internet and full control of your library data. Recommendation R5 in section 18 proposes a safer arrangement at no extra cost, and R4 still proposes dropping the Member / Admin / Super admin choice from the sign-in screen.

Branches and categories

New as a section in version 3.0. Two codes now sit inside every book number — the branch's two letters and the category's letter and digit — so both are defined here, ahead of the catalogue, with the rules that keep them stable.

ID	REQUIREMENT	PRIORITY
FR-BRN-01	NEW A super admin creates a branch with the library name, the branch name, a two-letter branch code, an address and a phone number.	MUST
FR-BRN-02	NEW The branch code is **exactly two capital English letters**, for example `KS`. The super admin chooses it, the system refuses one already in use, and it can be corrected only until the first book carrying it is accessioned. After that it never changes, because it is printed on the books.	MUST
FR-BRN-03	NEW A super admin can change a branch's name, and the library's name, at any time. The new name appears everywhere at once. Codes and accession numbers are untouched. The change is logged with the old and new names.	MUST
FR-BRN-04	NEW A super admin can upload a **logo** for each branch: a PNG, JPEG or WebP image up to 2 MB, which the system resizes and stores inside itself. The logo appears on the member's dashboard and the branch's pages, on staff screens, and on every PDF the system produces for that branch. A branch with no logo shows the library-wide logo, or the name in plain text if there is none.	MUST
FR-BRN-05	NEW A category has a code of **one capital letter followed by one digit**, for example `A1`, and a readable name. A super admin creates categories; the code is unique across the library and cannot be changed once any book carries it. The name can be changed at any time.	MUST
FR-BRN-06	NEW Members see category **names** wherever a category is shown, never the bare code. The code appears only inside the accession number and in staff screens.	MUST
FR-BRN-07	NEW A category can be retired so that no new book joins it, while the books already in it keep their numbers and remain in the catalogue.	SHOULD

> THE NEW CEILING
>
> Two letters give **676 branches**; a letter and a digit give **260 categories**; five digits give 99,999 books per branch per category. Version 2.0's 26-branch limit (risk D10) is gone. The categories are the number to watch now: 260 is plenty for a public library's own scheme, but it is not enough to mirror a standard classification such as Dewey, which was never the intention (risk D9).

06

The book catalogue

Every physical book on your shelves is one entry in the catalogue, identified by a number written or stuck on the book itself. The number changes shape again in this version, to make room for more branches and more categories.

The accession number

Eleven characters, always the same shape: `KS-A1-01234`.

KS

Branch code

Two capital letters, given to the branch when it is created. Unique across the library and never reused.

A1

Category code

One capital letter and one digit — the book's category, each with a readable name.

01234

Running number

Five digits, counting up from 00001 within that branch and that category. The system allocates it, not the librarian.

EXAMPLE	READS AS
KS-A1-00001	Branch KS, category A1, the first book of that category accessioned at that branch.
KS-A1-00100	Branch KS, category A1, the hundredth.
KS-A2-00001	Branch KS, category A2 — a different counter, so this is also a first book.
GL-A1-00001	Branch GL, category A1. Numbered on the same day as KS-A1-00001 and still unique.

ID	REQUIREMENT	PRIORITY
FR-CAT-01	CHANGED An admin or a super admin registers a book with its title, category, writer, publisher, how the library obtained it (purchase, donation, or the donor's name), and optionally its ISBN and its price in rupees. The price is **no longer required**. The accession number is produced by the system.	MUST
FR-CAT-02	CHANGED The accession number is **branch code, category code and five digits, separated by hyphens**, for example `KS-A1-01234`. The system allocates the next free number in that branch and category automatically, rejects any other format, and refuses a number already in use anywhere in the library. The hyphens are part of the number as printed; the system accepts it typed with or without them.	MUST
FR-CAT-03	Each book belongs to exactly one branch, and its condition is recorded as Useable, Missing or Not useable.	MUST
FR-CAT-04	CHANGED Each book shows as On the shelf, On loan, **In transit** or **Held for a member** (the last two arise from other-branch requests, section 8). A book that is missing, not useable, in transit or held can never be issued to anyone else.	MUST
FR-CAT-05	A member sees their home branch's catalogue by default and can filter it by category, writer and availability.	MUST
FR-CAT-06	A member can switch the view to any other branch and see which branch holds a copy of a book.	MUST
FR-CAT-07	CHANGED A search box finds books by title, writer, accession number or ISBN, across one branch or all branches. Typing `KS` lists that branch's books; typing `KS-A1` lists that branch and category.	MUST
FR-CAT-08	An admin can edit any detail of a book held at their own branch. A super admin can edit any book and is the only role that can move a book to a different branch permanently.	MUST
FR-CAT-09	CHANGED Category codes such as `A1` each carry a readable name, so members see "Sinhala fiction" rather than "A1". The codes and names are defined in section 5.	MUST

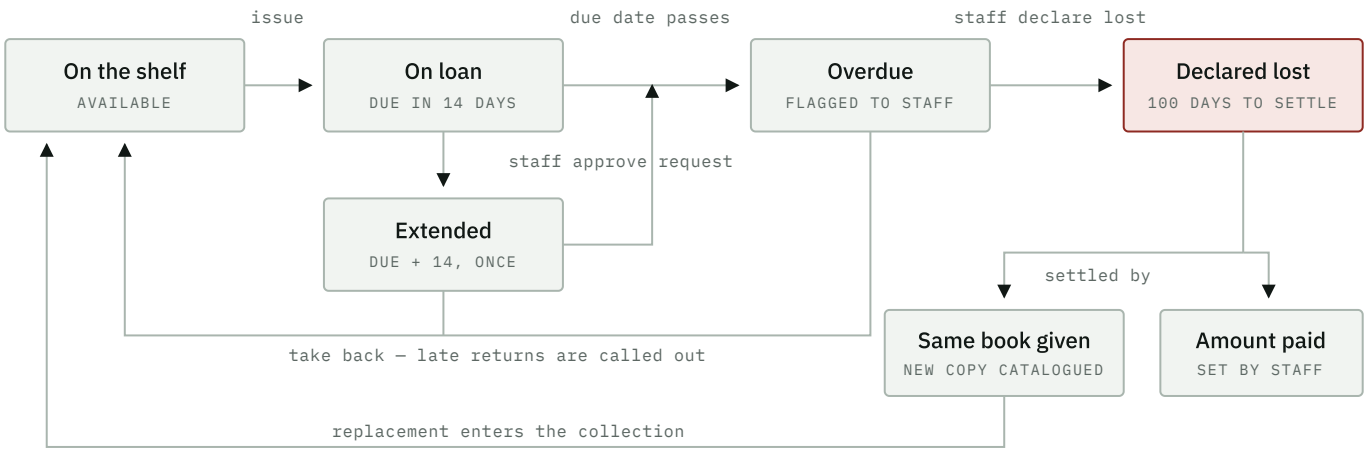
ID	REQUIREMENT	PRIORITY
FR-CAT-10	CHANGED The system can print a spine label carrying the accession number as a scannable barcode, in a format that carries the hyphens as printed.	COULD
FR-CAT-11	An admin cannot delete a book, or any other record. A book that leaves the collection is marked withdrawn with a reason and disappears from the catalogue while its history stays intact. Only a super admin can delete outright.	MUST
FR-CAT-12	CHANGED A book **may** carry a price in rupees. Where it does, the system uses it only to *suggest* a lost-book amount (FR-LST-02); where it does not, nothing is suggested and staff enter the amount. There is no longer a branch default price and no "needs a price" flag.	SHOULD
FR-CAT-13	Books already on the shelves with old-style numbers can be brought in by a one-off import that gives each one a new accession number and prints the new labels. The old number is kept alongside so an old card index still matches.	SHOULD
FR-CAT-14	An accession number, once given to a copy, never changes. If the book is later moved to another branch or its category is corrected, the number stays as it was printed on the book, and the branch and category the system works from are separate fields. The number records where a book was accessioned, not where it stands today.	MUST
FR-CAT-15	CHANGED Each branch is given its two-letter code when it is created, under the rules in FR-BRN-02.	MUST
FR-CAT-16	NEW When an ISBN is typed for a new book, the system can offer to fill in the title, writer and publisher from a free and open book-data source. This is a convenience only; cataloguing works exactly the same without it, and no paid service is involved.	COULD
FR-CAT-17	NEW The spine label can carry the accession number as a QR code as well as a barcode, so a phone camera can read it at the shelf.	COULD

> CARRIED OVER FROM VERSION 1.0
>
> Recommendation R1 in section 18 — storing the **title** once and the physical copies against it — is still open. It matters more in this version: an "identical copy" for a lost-book replacement (FR-LST-05) and "is there a copy at another branch" (section 8) are both questions the system can answer exactly only if it knows which copies are the same book.

07

Issuing, taking back and extending

This is the part of the system your staff will use every day, so it is designed around the counter: find the member, find the book, confirm. Two things are new: the member can ask for more time, and the system speaks up when a book comes back late.



The life of one book copy in version 3.0. New since v2.0: the loop through "Extended", the callout at take-back, the 100-day window, and the amount being set by staff rather than doubled from a price.

ID	REQUIREMENT	PRIORITY
FR-CIR-01	Staff issue a book by selecting a member and a book. The system records today as the borrow date and sets the due date 14 days later.	MUST
FR-CIR-02	CHANGED The system refuses to issue a book and states the reason if: the book is on loan, missing, not useable, withdrawn, in transit or held for another member; the member is still an applicant, or their membership has expired; the member already holds two books; the member has an open lost-book settlement and already holds one book; or a settlement has passed its 100-day window unsettled.	MUST
FR-CIR-03	Staff take a book back in one action. The return date is recorded and the book returns to the shelf immediately — or, for a book from another branch, goes into transit back (FR-REQ-07).	MUST
FR-CIR-04	CHANGED If a book comes back after its due date, the counter screen shows a **prominent notice** with the number of days late before the take-back is confirmed. The days late stay on the loan record, the member's page keeps a running count of late returns with the dates, and the member sees the same on their own dashboard the next time they sign in.	MUST
FR-CIR-05	Staff can correct a loan record — wrong book, wrong member, wrong date. A correction never overwrites history: the previous values and who changed them are kept in the activity log. An admin cannot delete a loan; only a super admin can.	MUST
FR-CIR-06	CHANGED A member may hold at most **two** books at a time, and at most **one** while a lost-book settlement is open (FR-LST-04). Both numbers are branch settings so they can be varied without new software.	MUST
FR-CIR-07	The loan period is a setting per branch, defaulting to 14 days, so a branch can run a longer or shorter period without new software. The extension period (FR-CIR-10) is the same setting.	SHOULD
FR-CIR-08	CHANGED At take-back, staff can declare the book lost or not useable instead of returning it to the shelf. Declaring it lost asks for the amount owed and starts the settlement described in section 9.	MUST
FR-CIR-09	A member's page shows their complete borrowing history, most recent first, including extensions and late returns.	SHOULD

ID	REQUIREMENT	PRIORITY
FR-CIR-10	CHANGED A member can **request one extension** on each loan from their dashboard, at any time before the due date. An admin of the home branch or a super admin approves or declines it, with a reason if declined. On approval the due date moves **14 days on from the original due date**, whatever day the approval happens. A loan can be extended once only, and not once it is overdue. Promoted from "could" to "must".	MUST
FR-CIR-11	NEW Staff can grant the same extension directly from the loan record when a member asks in person or by telephone, under the same rules. The log records whether the extension came from a member's request or a staff action.	MUST
FR-CIR-12	NEW Extension requests waiting for a decision appear in the staff notification panel. The member sees the request as pending, approved or declined on their dashboard, with the new due date or the reason.	MUST
FR-CIR-13	NEW A request still undecided two days before the due date is shown as urgent to staff. If it is still undecided when the due date passes, the loan becomes overdue as normal — a request is not an extension until someone approves it. Question Q3 in section 21 asks whether you would rather the clock paused.	SHOULD

> READING YOUR ITEM 4
>
> "If a user does not return a book, it must be noticed when the user returns it." We have read that as two things: the librarian must not be able to take back a late book without seeing that it was late (FR-CIR-04), and the lateness must follow the member — on the counter screen whenever staff open their record (FR-NOT-11), and on their own dashboard. If you meant something else, for instance a message the member must acknowledge before borrowing again, question Q4 in section 21 is where to say so.

08

Books from other branches

New in version 3.0. A member borrows only at their home branch. When the book they want is on the shelf at another branch, they ask for it, staff at both branches see who is asking, and the book travels — there, and back again after it is returned.

`Requested` → `Accepted` → `In transit to home branch` → `Held for member` → `On loan` → `Returned` → `In transit back` → `On the shelf`

ID	REQUIREMENT	PRIORITY
FR-REQ-01	NEW A member can request any book that is on the shelf at another branch, from the book's page in the catalogue. The request goes to the branch holding the book and is also visible to the member's home branch and to super admins.	MUST

ID	REQUIREMENT	PRIORITY
FR-REQ-02	NEW The request is accepted only if the member could be issued a book right now under FR-CIR-02, and the member may have at most one request open at a time. Applicants cannot request. The reason is stated on screen when a request is refused.	MUST
FR-REQ-03	NEW Staff looking at a request see the member's **full details**: full name, WhatsApp number, other phone number, address, date of birth and age, organisation, the person who introduced them, home branch, membership expiry date, the books they currently hold, any open settlement, and their count of late returns.	MUST
FR-REQ-04	NEW Staff at the holding branch accept or decline the request, with a reason if declined. Accepting takes the book off the shelf and marks it **in transit** to the member's home branch; from that moment it cannot be issued to anyone. The member sees the status change on their dashboard.	MUST
FR-REQ-05	NEW Staff at the home branch record the book's arrival. It is then **held for the member** for seven days (a branch setting), the member is told on their dashboard that it is ready, and it is issued to them at the counter as an ordinary loan with the ordinary due date.	MUST
FR-REQ-06	NEW If the member does not collect within the hold period, the hold lapses, staff are told, and the book is sent back: staff record dispatch at the home branch and arrival at the owning branch, where it returns to the shelf.	MUST
FR-REQ-07	NEW When the member returns the book, it is taken back at the home branch like any other loan and immediately marked in transit back to its owning branch; arrival there puts it on the shelf. The book's owning branch never changes through any of this, and its accession number is untouched.	MUST
FR-REQ-08	NEW The catalogue shows a travelling book as In transit or Held, never as On the shelf, at both branches.	MUST
FR-REQ-09	NEW The staff notification panel lists, for each branch: requests awaiting a decision, accepted books not yet dispatched, books in transit to or from the branch, and held books whose hold lapses within two days.	MUST
FR-REQ-10	NEW Every step — request, accept, decline, dispatch, arrival, hold lapse, return dispatch — is written to the activity log with who did it and when, so a book that goes missing between branches can be traced to the last hand it was in.	MUST
FR-REQ-11	NEW A member can cancel their own request until it is accepted.	SHOULD
FR-REQ-12	NEW A book in transit for more than a set number of days (default seven) is flagged to both branches and to super admins.	SHOULD

> **WHAT THE SYSTEM DOES NOT DO HERE**
>
> It does not carry the book. Somebody has to take it from one branch to the other, and the system only records that it left and that it arrived. How often books move between branches, who carries them, and who is responsible if one is lost on the way are library procedures, and risk D15 in section 19 is about what happens if they are not written down. Question Q8 in section 21 asks for them.

Lost and damaged books

A member who loses a book has two ways to put it right: bring the same book again, or pay the amount the librarian sets. They have 100 days to do one of them. In the meantime they may borrow one book instead of two.

ID	REQUIREMENT	PRIORITY
FR-LST-01	Staff can declare a book lost, either at take-back or later from the loan record, giving a short reason. The copy's condition becomes Missing and it leaves circulation at once.	MUST
FR-LST-02	CHANGED Declaring a book lost creates a **settlement** owed by the member who held it. The admin or super admin declaring it **sets the amount** in rupees. If the book has a recorded price the system suggests twice that price; the staff member may accept or change the figure. The amount, who set it and the suggestion it replaced are all kept and logged. The settlement offers two ways to clear it, and no others: an identical copy of the same book, or payment of that amount.	MUST
FR-LST-03	CHANGED The amount due and the **days remaining** in the window are shown on the member's page, on the staff screen and in the member's own dashboard, so nobody has to calculate either at the counter.	MUST
FR-LST-04	CHANGED While a settlement is open and within its window, the member may hold at most **one** book on loan; the lost copy itself does not count. Once the window has passed unsettled, the member cannot be issued any book, cannot request from another branch and cannot extend, and the reason is stated on screen when staff try.	MUST
FR-LST-05	An identical copy means the same title, writer and publisher, in useable condition. Staff confirm the match at the counter; the system records which book was accepted as the replacement.	MUST
FR-LST-06	A replacement copy is catalogued as a **new copy with a new accession number** for the branch and category it is joining. The lost copy stays lost and written off. The old number is never reissued.	MUST
FR-LST-07	A payment settlement records the amount, the date, a receipt number and the staff member who took it. Payments are never edited or deleted; a mistake is corrected by a reversing entry, so that both figures stay visible.	MUST
FR-LST-08	Lost-book money is kept separate from annual fee money, so a fee collection report is not inflated by replacement charges.	MUST
FR-LST-09	Only a super admin can waive a settlement or reduce the amount after it is set, and only with a written reason. The original amount, the new amount and the reason are all kept and shown in the activity log.	SHOULD
FR-LST-10	CHANGED The staff dashboard and the reports both list unsettled lost books, with the member, the book, the date declared, the amount due, the days remaining, and whether the window has passed.	MUST
FR-LST-11	A book that comes back damaged but still readable can be marked Not useable without a settlement, at staff discretion. A book damaged beyond use is treated exactly as a lost book.	SHOULD
FR-LST-12	If a book declared lost is later found, a super admin can return it to the shelf. If a settlement was already paid, the system shows it alongside so the library can decide about a refund; the refund itself is a decision for staff, not an automatic reversal.	SHOULD
FR-LST-13	A member's page shows every settlement they have ever had, open and cleared, with how each was cleared and how long it took.	SHOULD
FR-LST-14	NEW A settlement has a **window of 100 days** from the day the book was declared lost, held as a branch setting. The member's dashboard counts the days down; the staff panel shows windows closing within 14 days and windows already passed.	MUST

ID	REQUIREMENT	PRIORITY
FR-LST-15	NEW When the window passes with the settlement still open, the member is blocked as in FR-LST-04, the settlement is marked overdue, and it appears on every super admin's panel. Nothing else happens automatically: the membership is not cancelled and the amount does not increase unless you tell us otherwise. Question Q1 in section 21.	MUST

WHAT CHANGED, AND WHY IT NEEDS CARE

In version 2.0 the charge was a formula — twice the recorded price — and so it was the same whoever was at the counter. In this version it is a **judgement by the librarian**, which is fairer for a donated book with a 1990 price on it, and less fair between two members charged different amounts for the same book by two different librarians. The suggestion in FR-LST-02, the log, and the super admin's power in FR-LST-09 are the counterweights. Risk D14 in section 19 says more, and recommendation R17 suggests a written tariff.

10

Membership and annual fee

ID	REQUIREMENT	PRIORITY
FR-MEM-01	CHANGED A membership is valid for one year. The first year runs from the day the application is **approved**, or from the day a member is registered at the counter by staff — never from the day the public form was filled in.	MUST
FR-MEM-02	Staff record an annual fee payment against a member. The membership is extended by one year from its previous expiry date, so a member who pays late does not gain extra time.	MUST
FR-MEM-03	Staff see three lists on one screen: paid in the last 30 days, expiring in the next 30 days, and already expired.	MUST
FR-MEM-04	An expired member can still sign in and browse the catalogue but cannot be issued a book until the fee is paid.	MUST
FR-MEM-05	Each payment is kept as its own record with the amount, the date, a receipt number and the staff member who collected it. Payments are never overwritten and never deleted by an admin.	SHOULD
FR-MEM-06	CHANGED The annual fee amount is a setting per branch that **only a super admin** can change. A change applies to payments recorded from then on, never to payments already taken, and is logged with the old and new amounts.	MUST
FR-MEM-07	The system can print or export a receipt for a payment.	COULD
FR-MEM-08	CHANGED A member's record holds their date of birth. It is required on every new application and optional only for members registered before this version.	MUST

ID	REQUIREMENT	PRIORITY
FR-MEM-09	NEW A member's record holds every field from FR-AUT-03 — including the other phone number, organisation and the person who introduced them — and all of it is shown on the member's page to staff and to the member themselves.	MUST

11

Opening hours

ID	REQUIREMENT	PRIORITY
FR-SCH-01	Staff set a weekly pattern for their branch: for each day of the week, an opening and closing time, or closed.	MUST
FR-SCH-02	Staff can mark specific dates as closed with a reason, for Poya days, public holidays and stock-taking. These override the weekly pattern.	MUST
FR-SCH-03	A member sees their home branch's hours on their dashboard and can look up any other branch's hours.	MUST
FR-SCH-04	Today's status is stated plainly: Open until 5.00 pm, Opens at 9.00 am, or Closed today, with the reason where one is recorded.	SHOULD
FR-SCH-05	A closure can be set for a date range in one action rather than day by day.	COULD

12

Notices and events

New in version 3.0. A librarian can put a notice or an event in front of members: a title, a picture and a paragraph. Members see them on their dashboard alongside their branch's opening hours.

ID	REQUIREMENT	PRIORITY
FR-PUB-01	NEW An admin or super admin can publish a **notice** with a title, an optional image, and a paragraph of text that may contain a web link. An admin publishes for their own branch; a super admin can publish for one branch or for every branch at once.	MUST
FR-PUB-02	NEW An **event** has the same fields as a notice plus a date, a start time, and optionally an end time and a place. Events are listed soonest first, are marked as past once their date has gone, and drop off the member's dashboard seven days after that.	MUST
FR-PUB-03	NEW A notice can be given a date from which it is shown and a date after which it is hidden. A hidden or withdrawn notice is kept, not deleted, and can be shown again.	SHOULD

ID	REQUIREMENT	PRIORITY
FR-PUB-04	NEW A member's dashboard shows the notices and events of their home branch and those published for every branch, newest first. A member can look at another branch's notices and events, as they can its opening hours. Applicants see them too.	MUST
FR-PUB-05	NEW An image is a PNG, JPEG or WebP up to 2 MB. The system resizes it on upload, stores it **inside the system**, and shows a small version in the list and the full version when opened. No image is ever fetched from, or stored on, an outside website.	MUST
FR-PUB-06	NEW An admin can edit or withdraw notices and events of their own branch; a super admin any. Publishing, editing and withdrawing are logged. Only a super admin can delete one outright, and then its image goes with it.	MUST
FR-PUB-07	NEW A notice can be marked important, in which case it also appears in the member's notification panel until they have opened it.	COULD

WHY THE PICTURES LIVE INSIDE THE SYSTEM

You asked whether images could be "hard-coded or kept on a free service". A free image-hosting site is a paid service with the price set to zero for now: its terms change, its free tier shrinks, and every picture on every notice breaks on the same day. That is the situation NFR-11 exists to prevent. Storing the pictures in Pothgula's own database costs nothing, needs no account anywhere, and is backed up with everything else. The one price is size, which is why FR-PUB-05 resizes on upload: a notice picture does not need to be a 4 MB photograph. Risk D18 in section 19 puts numbers on this against the free database tier proposed in the technical decision record.

13

Alerts and reminders

Every alert this system produces is shown on the dashboard of the person who needs it. There is no other channel: nothing is sent by SMS, WhatsApp, email or push notification, because every one of those costs money to run.

ID	REQUIREMENT	PRIORITY
FR-NOT-01	A member's dashboard warns them three days before a book is due, and shows a stronger warning every day it stays overdue. *Question Q7 asks whether this should become five days to match the staff list.*	MUST
FR-NOT-02	A member is warned 30 days before their membership expires, and told clearly once it has expired.	MUST
FR-NOT-03	CHANGED The staff dashboard opens on today's work: books due back today, books due within five days, books already overdue, applications waiting for approval, extension requests, other-branch requests needing action, memberships expiring this month, and lost books not yet settled.	MUST

ID	REQUIREMENT	PRIORITY
FR-NOT-04	Staff can produce a call list of overdue members with their phone numbers, for following up by telephone or WhatsApp by hand.	SHOULD
~~FR-NOT-05~~	WITHDRAWN The system sends an SMS reminder for a due book and an expiring membership. *Removed in v2.0; still removed.*	REMOVED
FR-NOT-06	Every alert and reminder in the system is delivered by showing it in the dashboard of the signed-in person concerned. No feature depends on a message leaving the system.	MUST
FR-NOT-07	Each dashboard opens with a notification panel listing everything currently needing that person's attention, most urgent first, with a count. An item disappears from the panel as soon as the thing that caused it is dealt with.	MUST
FR-NOT-08	CHANGED A member's panel covers: books due within three days; overdue books; late returns on their record; an extension request decided; an other-branch request accepted, declined or ready to collect; an open lost-book settlement with the amount and the days remaining; membership expiring within 30 days or already expired; closures at their home branch in the next seven days; and, for an applicant, that the application is waiting.	MUST
FR-NOT-09	CHANGED A staff panel covers: books due back today; the five-day list of FR-NOT-12; overdue books; late returns taken back today; applications waiting for approval; extension requests waiting; other-branch requests at each step (FR-REQ-09); unsettled lost books, with windows closing within 14 days and windows already passed; and memberships expiring this month. An admin sees their own branch; a super admin sees every branch, grouped by branch.	MUST
FR-NOT-10	Alerts are worked out from live data each time a dashboard is opened, not stored and queued, so a notification can never be stale or arrive after the thing it warns about has been dealt with.	SHOULD
FR-NOT-11	CHANGED When staff open a member's record at the counter, anything overdue, any late return in the last year, any open settlement and any pending application is shown at the top of the page before anything else. Promoted from "should" to "must".	MUST
FR-NOT-12	NEW The **five-day list**: every loan due within the next five days and not yet returned, showing the member's full name, WhatsApp number and other phone, their home branch; the book's title, accession number and owning branch; the due date; and whether an extension has been requested or granted. An admin sees their own branch's loans; a super admin sees all, grouped by branch. The number of days is a setting.	MUST

> **A WORD ON THE FIVE-DAY LIST**
>
> With a 14-day loan, a book spends more than a third of its life "due within five days". At two hundred loans a week the list will hold seventy or eighty lines every morning, most of them books that will come back on time. We have written it as a list on the dashboard rather than as an alert per book, so that it does not bury the things that are actually wrong — the overdue books and the passed settlement windows. Recommendation R16 in section 18 says a little more.

THE CONSEQUENCE, STATED PLAINLY

A member who never signs in will never see a reminder. Those are exactly the members who go overdue. The answer has to be a human one: the call list in FR-NOT-04, and staff telephoning or messaging people by hand. Having the WhatsApp number as the sign-in name makes that easier for the person making the calls; it does not let the system send anything. Risk D6 in section 19 puts numbers around what that means for your staff.

14

The activity log

Everything anybody does in this system that changes anything is written down, permanently, and both admins and super admins can read it and take it away as a file.

ID	REQUIREMENT	PRIORITY
FR-LOG-01	Every action that creates, changes or removes data is written to the activity log, whoever performed it — applicant, member, admin or super admin.	MUST
FR-LOG-02	Each entry records the date and time, the account that acted and its role, the branch, what was done, which record it was done to, the values before and after the change, and the network address of the device used.	MUST
FR-LOG-03	CHANGED Member actions written to the log: signing in, a failed sign-in attempt, signing out, applying, changing a password, editing their own details, requesting or cancelling an extension, and requesting or cancelling a book from another branch.	MUST
FR-LOG-04	CHANGED Staff actions written to the log: everything in FR-LOG-03, plus approving or rejecting an application; exporting a member's details; issuing a book, taking one back, correcting a loan; approving or declining an extension; every step of an other-branch request; adding, editing or withdrawing a book; registering or deactivating a member; taking an annual fee; declaring a book lost and setting the amount; settling or waiving a lost book; changing opening hours or closures; publishing, editing or withdrawing a notice or event; resetting another person's password; creating or editing a branch, its name or its logo; creating or retiring a category; changing the annual fee or any other branch setting; creating a staff account; and exporting the log itself.	MUST
FR-LOG-05	The log is append-only. Nobody — not an admin, not a super admin, not us — can edit or delete an entry from inside the application.	MUST
FR-LOG-06	An admin can read the log for their own branch. A super admin can read the log for every branch.	MUST
FR-LOG-07	The log screen filters by date range, person, role, branch, kind of action and the record affected, and shows the newest first.	MUST
FR-LOG-08	CHANGED Any log view can be exported as a **plain text file**, as a spreadsheet file and as a **PDF**, all three produced by the system itself with no outside service. The PDF carries the branch name and logo, the filter used and the date it was produced. Promoted from "should" to "must".	MUST
FR-LOG-09	Every record — a book, a member, a loan, a payment, a request, a notice — has a "history" tab showing its own log entries in order.	SHOULD

ID	REQUIREMENT	PRIORITY
FR-LOG-10	Reading the log is itself logged, so it is visible who has been looking at whose activity.	SHOULD
FR-LOG-11	Log entries are kept for a retention period held as a setting, at least two years by default. Entries older than that are moved to an archive file, never destroyed.	SHOULD
FR-LOG-12	Reads are not logged. Browsing the catalogue, running a search or opening a book's page produces no entry, because doing so would multiply the size of the log many times over for no benefit and would create a detailed record of what every member has been reading.	MUST

15

Reports and exports

Summaries were optional in version 1.0 and core in version 2.0. This version adds the exports you asked for — text and PDF everywhere — and two reports that follow from the new features.

ID	REQUIREMENT	PRIORITY
FR-RPT-01	Stock summary for a branch: how many books in each category, and how many are useable, missing, not useable, in transit or held.	SHOULD
FR-RPT-02	Currently issued and overdue lists, with extensions shown.	SHOULD
FR-RPT-03	Fee collection summary for a chosen date range, per branch and in total, with lost-book charges shown separately.	SHOULD
FR-RPT-04	Membership count over time: applications, approvals, rejections, renewals and lapsed members per month.	SHOULD
FR-RPT-05	**Most borrowed and least borrowed books** over any chosen period, with the number of times each was taken out, for one branch or the whole library, and filterable by category.	MUST
FR-RPT-06	An at-a-glance summary screen showing, in one view: total books held, how many are on loan now, how many are overdue now, active and expired members and waiting applications, unsettled lost books, fees collected this month, and the most borrowed book this month.	MUST
FR-RPT-07	A missing-books report listing every book declared lost or not useable in full detail — accession number, title, category, branch, the member who last held it, the date it was declared, the amount set, and whether the settlement is open, cleared or past its window.	MUST
FR-RPT-08	Borrowing by category and by month, so the library can see what is read and when the busy months are.	SHOULD
FR-RPT-09	CHANGED Every report can be exported as a **plain text file**, a spreadsheet file and a **PDF**, generated by the system with no paid service and no external website involved. Promoted from "should" to "must".	MUST
FR-RPT-10	An admin sees reports for their own branch. A super admin sees any branch and the whole library combined, with branch-by-branch comparison.	MUST
FR-RPT-11	Membership by age band, drawn from the date of birth, so the library can see who it is serving.	COULD
FR-RPT-12	A year-end summary gathering the reports above into one printable document.	COULD

ID	REQUIREMENT	PRIORITY
FR-RPT-13	NEW A **single member's details** — every field on their record, their membership dates, current loans and open settlements — exported as a text file or a PDF, as FR-AUT-19 describes. This is the only export that carries one person's personal data on its own, and it is logged each time.	MUST
FR-RPT-14	NEW Other-branch requests over a period: how many, between which branches, how many declined and why, and how long each took from request to collection.	SHOULD
FR-RPT-15	NEW Late returns over a period: the members with the most late returns and the average days late, per branch.	SHOULD

> **"TEXT" AND "EXCEL"**
>
> Your notes say text and Excel; your list says text and PDF. We have written all three. By a text file we mean a plain `.txt` that opens anywhere and a `.csv` that opens in Excel or LibreOffice as a spreadsheet; the spreadsheet file itself is `.xlsx`. If you only want some of these, question Q11 in section 21 is where to say which.

16

Quality requirements

These apply to the whole system rather than to one feature, and they are as much a part of acceptance as the features above.

ID	AREA	REQUIREMENT
NFR-01	Speed	Any counter action — search, issue, take back — completes in under two seconds with 50 people using the system at once.
NFR-02	Devices	Works on a desktop browser at 1366×768 and on a phone. Members will mostly use phones; staff will mostly use the counter machine.
NFR-03	Browsers	Current versions of Chrome, Edge, Firefox and Safari.
NFR-04	Language	Sinhala, Tamil and English text can be entered and displayed correctly in every field, including names, book titles, notices and PDF exports.
NFR-05	Time	All dates and times are shown in Sri Lanka time. Due dates, extension dates and the 100-day window are calculated by date, not by hour.
NFR-06	Security	All traffic over HTTPS. Passwords hashed with a modern algorithm. Staff pages unreachable without a valid staff session, checked on the server for every request — including the branch a staff member is allowed to act on. Uploaded images are checked to be images and are never served in a way that lets them run as code.
NFR-07	Audit	Every action that changes data records who did it and when, and cannot be edited from within the application. Section 14 sets out what this covers in full.
NFR-08	Backup	The database, including the images in it, is backed up daily and a restore is tested before handover. Recovery point: at most 24 hours of data. Backups run on free, self-hosted tooling.

ID	AREA	REQUIREMENT
NFR-09	Privacy	CHANGED Member names, WhatsApp and other phone numbers, dates of birth, addresses, organisations and the names of the people who introduced them are personal data under Sri Lanka's Personal Data Protection Act No. 9 of 2022 — and so are the same details of applicants who are never approved. Access is limited to staff of the member's own branch, to staff of a branch handling a request from that member, and to super admins. The activity log and every exported file are covered by the same restriction.
NFR-10	Correctness	The same book cannot be issued twice at the same moment, cannot be issued while in transit or held, and two branches cannot be given the same accession number, branch code or category code. All are enforced in the database, not only in the screen.
NFR-11	Cost	No part of the system may depend on a paid third-party service, or on any charge per message, per request, per user or per month. This applies to reminders, password resets, maps, PDF generation, image storage, fonts, error reporting and analytics alike.
NFR-12	Licence	Every framework, database, library and tool used is free and open-source software under a permissive licence, and the library owns and can modify its own copy of the system without asking anyone.
NFR-13	Hosting	The whole system can be installed and run by the library on its own machine or on a low-cost server, with written setup instructions and no vendor account required to run it.
NFR-14	Money	All amounts are held in whole cents of Sri Lankan rupees as exact numbers, never as approximate decimals, so no total can drift by a cent.
NFR-15	Phone numbers	CHANGED Phone numbers are held as text, never as numbers, so a leading zero survives. The WhatsApp number is unique because it is also the sign-in name; the other phone number need not be.
NFR-16	Images	NEW Every image — logos, notice and event pictures — is stored inside the system's own database, resized on upload to a bounded size, limited to PNG, JPEG and WebP, and served only by the system. No image is hosted outside it and no upload is accepted from a web address.

17

Information we store

This is the record structure behind the system. It is included so you can confirm nothing you need is missing before we build. Fields shown in green are new or changed in version 3.0.

branch

id	number
library_name	text
branch_name	text
branch_code	two letters, unique, fixed
logo_image_id	image, optional
address	text
phone	text
created_at / updated_at	date

category

code	letter + digit, unique, fixed
name	Sinhala fiction, etc.
is_active	yes / no

book

accession_no	KS-A1-01234, unique
title	text
category_code	category
writer	text, optional
publisher	text, optional
isbn	text, optional
price_cents	LKR, optional
source	text, optional
status	useable / missing / not useable
location	shelf / transit / held
withdrawn_at	date, optional
branch_id	owning branch
accessioned_branch_id	branch that numbered it
legacy_no	old number, optional
created_at / updated_at	date

user

id	number
full_name	text
user_type	member / admin / super admin
status	pending / active / rejected / deactivated
whatsapp_phone	text, unique, sign-in
other_phone	text, optional
password	hashed
must_change_password	yes / no
date_of_birth	date
address	text
organisation	text, optional
introduced_by_name	text
branch_id	home branch
applied_at	date
approved_at / approved_by	date, staff user
rejection_reason	text, optional
expire_at	date
late_return_count	number, derived
created_at / updated_at	date

borrow

id	number
book_id	book
user_id	member
branch_id	branch issued at
request_id	other-branch request, optional
borrow_date	date
due_date	borrow + 14 days
original_due_date	date
extension_status	none / requested / approved / decl
extension_requested_at	date, optional
extension_decided_by / at	staff user, date
extension_reason	text, if declined
returned_at	date, empty while on loan
days_late	number, 0 if on time
lost_declared_at / by	date, staff user
issued_by / received_by	staff user
created_at / updated_at	date

book_request

id	number
book_id	book
user_id	member
from_branch_id	owning branch
to_branch_id	member's home branch
status	requested / accepted / in transit / held / is
requested_at	date
decided_by / decided_at	staff user, date
decline_reason	text, optional
dispatched_at / arrived_at	date
hold_until	date
return_dispatched_at / return_arrived_at	date

book_settlement

id	number
borrow_id	loan
book_id	the lost copy
user_id	member
suggested_cents	price × 2, optional
amount_due_cents	set by staff
amount_set_by	staff user
window_ends	declared + 100 days
cleared_by	replacement / payment / waiver
replacement_book_id	new copy, optional
amount_paid_cents	LKR, optional
receipt_no	text, optional
reason	text, for a waiver
settled_on	date, empty while open
handled_by	staff user

membership_payment

id	number
user_id	member
amount_cents	LKR, integer
paid_on	date
receipt_no	text
collected_by	staff user
extends_to	new expiry

notice

id	number
kind	notice / event
branch_id	branch, empty = all
title	text
body	text, may hold a link
image_id	image, optional
event_date / start_time / end_time	event only
venue	text, optional
show_from / show_until	date, optional
is_important	yes / no
withdrawn_at	date, optional
created_by	staff user
created_at / updated_at	date

image

id	number
data	bytes, resized
thumbnail	bytes
mime_type	png / jpeg / webp
width / height / bytes	numbers
uploaded_by	staff user
created_at	date

branch_schedule

branch_id	branch
weekday	Mon–Sun
open_time	time, empty = closed
close_time	time

branch_closure

branch_id	branch
closed_on	date
reason	text

branch_setting

branch_id	branch
loan_days	default 14
max_books	default 2
max_books_while_settling	default 1
settlement_window_days	default 100
staff_due_warning_days	default 5
member_due_warning_days	default 3
hold_days	default 7
application_stale_days	default 30
annual_fee_cents	LKR, super admin only
default_price_cents	removed
log_retention_months	default 24

activity_log

id	number
at	date and time, UTC
actor_id	any user
actor_role	member / admin / super admin
branch_id	branch
action	issued / approved / exported…
entity	book / borrow / user / request…
entity_id	record affected
before / after	values changed
ip_address	text

Three tables are new in this version. `book_request` follows a book from one branch to another and back, and is the only place the system knows a book is somewhere between two shelves. `notice` holds notices and events together, told apart by `kind`. `image` holds every picture — logos and notice images alike — in one place, so that there is one size limit, one backup and one thing to watch. The loan record grows fields for the single extension and the days late; the member record grows the application fields and the pending / active / rejected state; and the settlement now records the amount staff set alongside what the system suggested.

18

What we recommend changing

These are our professional recommendations on the design as it stands. None is a defect in your thinking — they are the places where a decision that is reasonable today becomes expensive in two years. Three more from earlier versions are settled by this update; four are new.

ID	RECOMMENDATION	WHY IT MATTERS	STATUS
R1	Separate the **title** from the **copy**.	Five copies of the same book mean typing the writer, publisher and title five times, and a typo in one splits the book into two entries in search results. In this version it also decides whether "an identical copy" for a lost book and "is there a copy at another branch" are things the system can check or things staff must eyeball.	Open
R2	Work out availability from the loans and requests rather than storing it as a separate yes/no field.	Two places recording the same fact will eventually disagree. With books now in transit as well as on loan, there are more ways for a stored flag to be wrong. One source of truth removes the whole class of problem.	Open

ID	RECOMMENDATION	WHY IT MATTERS	STATUS
R3	Store phone numbers as **text**, not as numbers, and require the WhatsApp number to be unique.	Stored as a number, `0771234567` loses its leading zero. Since the WhatsApp number is also the sign-in name, duplicates must be impossible.	Adopted — NFR-15
R4	Drop the Member / Admin / Super admin choice from the **sign-in** screen; keep it only for registration.	The password already determines who someone is. Asking first only creates a way to fail, and confirms to a stranger that a given number belongs to an admin.	Open
R5	Make the super admin code **single-use**. After the first super admin exists, further staff are created by an existing super admin, not by self-registration.	A fixed code shared among staff spreads by WhatsApp and never expires. Anyone who ever sees it can create a super admin account for themselves, at any time, from anywhere. Now that members are approved one by one, it is odd that a super admin is the one account anyone can create unchecked.	Open
R6	Have staff **approve** a member's self-registration before the membership becomes active, and start the year from that approval.	Adopted in full: FR-AUT-16 to FR-AUT-18 and FR-MEM-01.	**Adopted — v3.0**
R7	Record a **due date** and a separate **returned date** on each loan.	One date field whose meaning changes makes "which books are overdue right now" impossible to answer reliably.	Adopted — v1.0
R8	Decide the **rule for other branches**.	Decided: a member borrows at home and requests from elsewhere; the book travels. Section 8.	**Adopted — v3.0**
R9	Make super admin an **organisation-wide** role, and admin the branch-scoped one.	Super admins now set branch names and logos, define categories, set every branch's fee and see every branch's requests, all of which are organisation-wide acts. Yet FR-AUT-04 still caps them at five *per branch*.	Open
R10	Add **barcode or QR scanning** at the counter once the basics are live.	The eleven-character accession number is fixed-length and unambiguous, and a USB scanner or a phone camera reads it faster than anyone can type it. Your notes mark this for research; FR-CAT-10 and FR-CAT-17 hold the place.	Open — phase 3
R11	**Decide the branch-letter ceiling now**.	Decided: two letters. Section 5.	**Adopted — v3.0**
R12	Implement "delete" as **withdraw** everywhere, even for super admins.	If a super admin can still remove a book outright, the loans and requests that pointed at it are orphaned and the reports that counted it silently change.	Open
R13	Write the activity log **in the same transaction** as the change it describes.	If the log is written afterwards, an interruption between the two leaves a change with no record.	Open
R14	Record the **phone number** of the person who introduced the member, not only their name.	A name alone is a note; a name with a number is someone the library can call when a member with a lost book stops answering. If the introducer is meant to vouch for the member, the number is what makes that real. If they are only a note, the name is enough — question Q5.	New in v3.0
R15	When the 100-day window passes, **suspend** rather than cancel.	Cancelling a membership loses the loan history and the settlement with it, and the member simply applies again under the other phone number. Blocking every borrowing feature while keeping the account, as FR-LST-15 does, keeps the debt attached to the person.	New in v3.0
R16	Keep the **five-day list** as a working list, not as alerts.	It will be the longest thing on the staff screen every day, and most of what is on it is fine. If every line were an alert, the overdue books — the ones that actually need a phone call — would be lost among them. A list that staff can sort by due date, with the overdue books held separately above it, keeps the panel honest.	New in v3.0

ID	RECOMMENDATION	WHY IT MATTERS	STATUS
R17	Write a **tariff** for lost-book amounts and give it to every librarian.	Letting staff set the amount is right for a donated book with a 1990 price on it. But two librarians will set two different amounts for the same book unless there is a written rule — a table by category, or "current shop price, or twice the recorded price, whichever is higher". The system suggests a figure; the tariff is what makes the suggestion consistent between branches.	New in v3.0

19

Limitations and risks

What this system will not do for you, and what could go wrong. We would rather you know now than discover it after handover. Two risks from version 2.0 are closed by this update; five are new.

Serious — please decide on these before we build

Anyone with the code becomes an owner D1

The super admin registration code is a single shared secret with no expiry. A staff member who leaves keeps the ability to create a super admin account, read every member's details, delete records and waive charges.

Mitigation: adopt R5. Otherwise, rotate the code whenever anyone leaves and accept the risk in writing.

Unverified members can borrow real books D2

Closed in v3.0. Self-registration now produces an application, and nothing can be borrowed until a librarian approves it (FR-AUT-16, FR-AUT-17).

Late is still free; only lost costs money D3

A member can now ask for two more weeks, which is the honest route. But a member who simply keeps a book for a year still owes nothing as long as they eventually bring it back, and nothing in the system stops them borrowing again meanwhile. Version 3.0 calls out the lateness at the counter, which helps; it does not put a price on it.

Mitigation: tell us your fine policy and we will cost it. Blocking further borrowing while a book is overdue costs almost nothing and would close most of this gap on its own. Question Q14.

The counter stops when the internet does D4

This is a web system with no offline mode. During an outage at a branch, staff cannot issue or take back books and must record transactions on paper, then enter them later.

Mitigation: a printed daily loans sheet as a fallback, plus an agreed procedure for entering a backlog. Offline-capable software is a much larger project.

Reminders reach only people who sign in — permanently D6

A member who never signs in never learns their book is due, that their extension was declined, that a requested book is waiting for them, or that the 100-day window is about to close. Making the WhatsApp number the sign-in name does not change this: the system still cannot send a WhatsApp message.

Mitigation: the call list in FR-NOT-04 and the five-day list in FR-NOT-12, and staff time on the telephone or WhatsApp by hand. Plan for someone to do this weekly.

The lost-book amount is now a judgement D14

Two members losing the same book at two branches can be charged two different amounts, decided by two different librarians, and the log will show it. That is a fairness complaint waiting to happen, and in a small community it will be made in person.

Mitigation: R17 — a written tariff. The system's suggestion (FR-LST-02) and the super admin's power to adjust (FR-LST-09) exist so that departures from the tariff are visible and correctable. Question Q2.

A book between two branches belongs to nobody D15

The system knows a book left branch GL and has not arrived at branch KS. It does not know who is carrying it, and if it never arrives there is no member to settle with — the loss falls on the library and the log can only say which staff member pressed Dispatch.

Mitigation: a written transport procedure — who carries, how often, signed for at each end — and FR-REQ-12 flagging anything in transit too long. Question Q8.

Worth knowing

No reservations for books on loan D5

A member can now request a book that is on the shelf at another branch. They still cannot queue for a book that is on loan, at any branch: they must keep checking, and whoever arrives first gets it.

Option: phase 3 feature; the request flow in section 8 is most of the machinery.

Books now move between branches — and staff must keep up D7

The transit and hold states in section 8 are only as accurate as the staff who record dispatch and arrival. A book that arrived but was never marked as arrived sits "in transit" for ever, and the member who asked for it is never told it is ready.

Mitigation: FR-REQ-09 puts every undone step on the panel of the branch that owes it; FR-REQ-12 flags the stuck ones to super admins.

Personal data brings legal duties, and there is more of it now D8

The record now holds two phone numbers, an organisation and the name of a third person who did not fill in the form. Staff at a second branch see all of it when a request arrives. Applicants who are rejected leave their details behind. Single-person exports (FR-RPT-13) put all of it in a file that can be carried anywhere.

Mitigation: NFR-09, FR-AUT-20's purge of unsuccessful applications, and logging every export. We will supply a privacy notice for the application form. The policy decisions are yours. Questions Q19 and Q20.

The catalogue is local, not standard D9

Books are catalogued with your own codes rather than a standard classification. Simpler to use, but not importable into a standard library system without manual work — and the accession number now bakes your codes into every label.

Option: fill in the ISBN where the book has one; FR-CAT-16 makes that quicker.

The numbering scheme caps you at 26 branches D10

Closed in v3.0. Two letters allow 676 branches and a letter and a digit allow 260 categories.

The log will become the biggest thing in the database D11

Logging every change means the log grows without limit. Requests, extensions and approvals add entries on top of loans. It will outgrow every other table within the first year — except, possibly, the images (D18).

Mitigation: FR-LOG-11 archives entries past the retention period, and FR-LOG-12 keeps browsing out entirely. Question Q20.

A locked-out super admin has no self-service way back D12

A super admin who forgets their password depends on another super admin to reset it.

Mitigation: FR-AUT-14 keeps two super admins alive. As a last resort we will supply a command run directly on the server, documented at handover.

Twice the price may not replace the book D13

Superseded in v3.0. The doubled price is now only a suggestion and staff set the amount. The risk has moved: see D14.

The five-day list is long by design D16

More than a third of all loans are always "due within five days". If staff learn to scroll past the list, they will scroll past the overdue books above it too.

Mitigation: R16. The number of days is a setting, so it can be shortened if the list proves more noise than help.

Applications that nobody approves D17

An applicant who fills in the form and is never approved has a working sign-in, sees the catalogue, and can never borrow. They will not know whether they were rejected or forgotten, and the library holds their details all the while.

Mitigation: FR-AUT-18 counts them on every staff panel; FR-AUT-20 flags them after 30 days and purges after 90. The number of days is yours to set.

Pictures are the one thing that can fill the free database D18

The technical decision record proposes a free database tier of 0.5 GB. Every record in this system put together will not reach a tenth of that. Images can: two hundred notices with an unresized 2 MB photograph each is 400 MB, and the daily backup carries all of it.

Mitigation: FR-PUB-05 and NFR-16 resize on upload to a bounded size — a notice picture at 1200 pixels wide is around 150 KB, which makes the same two hundred notices 30 MB. Withdrawn notices keep their images; a super admin's delete removes them. If notices are frequent, self-hosting on a real server (NFR-13) removes the ceiling altogether.

20

Delivery in three phases

We recommend putting a working counter system in your hands early rather than delivering everything at once. Each phase is usable on its own. The new features are placed where they depend least on what is not yet built.

1

The counter works

FIRST RELEASE

Branches with their two-letter codes, names and logos; categories with their codes and names; accounts and sign-in by WhatsApp number; membership applications with approval, the pending list and the single-member export; password change and counter reset; the book catalogue with the new accession numbers and search; issuing and taking back with due dates and the late-return callout; the five-day list and the overdue list on the staff dashboard; membership expiry — and the activity log, from the first day. The log has to be there from the start: the actions it did not record are gone. At the end of this phase you can stop using the paper register.

2

The library runs itself

SECOND RELEASE

Extensions by request and approval; other-branch requests with transit and hold; the lost-book settlement with staff-set amounts and the 100-day window; opening hours and closures; notices and events; annual fee payments with history and receipts; the member-facing dashboard with its full notification panel; the log viewing screen with text, spreadsheet and PDF export; and the reports of section 15 with the same exports.

3

Faster and further

ONCE PHASES 1 AND 2 ARE SETTLED IN USE

Barcode and QR scanning at the counter and the shelf, filling in book details from an ISBN, reservations for books on loan, fines for late returns if you decide on them, and the year-end summary document. SMS and WhatsApp messaging are not on this list — they are out of scope by decision, not deferred.

21

Decisions we need from you

Each of these changes what we build. Q1 to Q13 arise from this update; Q14 onwards are carried from version 2.0 unanswered. Q1, Q2, Q6 and Q13 are needed before development starts.

Q1 A lost book is not settled by day 100. We have written that the member is blocked from every borrowing feature and the case goes to the super admins, and nothing else changes automatically. Is that right, or should the membership be suspended, the amount increased, or something else?

New. FR-LST-15, R15.

Q2 When staff set the lost-book amount, should the system still suggest twice the recorded price where there is one? And will you write a tariff so that different librarians reach the same figure?

New. FR-LST-02, R17, risk D14.

Q3 Extensions: one per loan, requested before the due date, running from the original due date — is that the rule? If staff have not decided a request by the due date, should the book go overdue as normal, or should the clock pause until they decide? May staff extend a book that is already overdue?

New. FR-CIR-10, FR-CIR-13.

Q4 Your item 4 — "if the user does not return a book, it must be noticed when the user returns it". We have read it as the late-return callout at the counter and on the member's record and dashboard. Is that what you meant, or something more, such as a message the member must acknowledge?

New. FR-CIR-04, FR-NOT-11.

Q5 Who is the "organiser" on the application form — a guarantor who takes responsibility, a referee, or simply a note of who brought the person in? Should we record their phone number as well as their name?

New. FR-AUT-03, R14.

Q6 Is the first year's fee collected at the moment of approval, so that approving and taking the fee are one step at the counter? Can a person be approved without paying?

New. FR-AUT-17, FR-MEM-01.

Q7 Staff are warned five days before a book is due. Members are warned three days before, as in version 2.0. Should the member's warning also become five days?

New. FR-NOT-01, FR-NOT-12.

Q8 How will books physically travel between branches — who carries them, how often, and who is responsible if one is lost on the way?

New. Section 8, risk D15.

Q9 We have assumed a member may request only a book that is on the shelf at the other branch, not one that is on loan there, and may have one open request at a time. Is that right?

New. FR-REQ-01, FR-

REQ-02.

Q10 Should members see only their own branch's notices and events by default, with other branches on request — as we have written — or every branch's at once?

New. FR-PUB-04.

Q11 By "text" export do you mean a plain `.txt`, a `.csv` that opens in Excel, or both? Your notes say text and Excel; your list says text and PDF. We have written all three.

New. FR-LOG-08, FR-RPT-09.

Q12 A requested book is held for the member for seven days after it arrives. Is that the right length?

New. FR-REQ-05.

Q13 What are the two-letter codes for each branch, and what are your categories with their letter-and-digit codes and readable names? Nothing can be catalogued until both lists exist, because both are inside every book number.

New, replacing v2.0 Q3 and Q10. Section 5.

Q14 Is there a fine for a book that is merely late rather than lost? If so, what is the daily rate and is there a maximum? Should a member with an overdue book be blocked from borrowing again?

Carried from v2.0 Q4. Risk D3.

Q15 What is the annual membership fee at each branch?

Carried from v2.0 Q6.

Q16 How many books and members in total, roughly, and how many notices a month? This decides how the reports, the log and the images are built.

Carried from v2.0 Q7.

Q17 Who will be the first super admin, who is the second required by FR-AUT-14, and who holds the setup code?

Carried from v2.0 Q8.

Q18 Should the interface be in English only, or English with Sinhala and Tamil?

Carried from v2.0 Q9.

Q19 Should member browsing and searching be logged as well as actions that change data? We recommend not.

Carried from

v2.0 Q12.

Q20 How long should activity log entries be kept before archiving, and how long should unsuccessful applications be kept? We have assumed two years and 90 days.

Carried from v2.0 Q13, extended. FR-AUT-20, FR-LOG-11.

Q21 Which open-source licence should the project carry, and who holds the copyright — the library, or you personally?

Carried from v2.0 Q14.

Q22 Where will the system be hosted, and who will run the daily backup and test the restore? The answer also decides whether the image ceiling in risk D18 matters.

Carried from v2.0 Q15.

22

Acceptance

Signing below confirms that the requirements marked `MUST` in sections 4 to 16 describe the system you want built, that the changes listed in section 2 are what you asked for, that the limitations in section 19 have been read and accepted, and that the answers to section 21 have been provided in writing. Anything not listed in this document is outside the agreed scope and will be quoted separately. This document supersedes version 2.0 of 8 September 2026 and version 1.0 of 25 August 2026 in full.

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Questions on any part of this document are welcome before signing. It is easier to change a requirement now than a screen later — and easier to change a screen than a number printed on four thousand books.